SOFTWARE TESTING ASSIGNMENT

Module 2 – manual Testing

1. What is Exploratory Testing ?

- Exploratory testing is an approach to software testing that is often described as simultaneous learning, test design , and execution.

2. What is traceability matrix ?

- A Traceability Matrix is a document that co-relates any two-baseline documents that require a many-to-many relationship to check the completeness of the relationship.
- It is used to track the requirements and to check the current project requirements are met.

3. What is Boundary value testing ?

- Software testing technique in which tests are designed to include representatives of boundary values. It is performed by the QA testing teams.

4. What is Equivalence partitioning testing ?

- In this method the input domain data is divided into different equivalence data classes. This method is typically used to reduce the total number of test cases to a finite set of testable test cases, still covering maximum requirements. In short it is the process of taking all possible test cases and placing them into classes. One test value is picked from each class while testing.

  - One input data class with all valid inputs. Pick a single value from range 1 to 1000 as a valid test case. If you select other values between 1 and 1000 then result is

going to be same. So one test case for valid input data should be sufficient.

- Input data class with all values below lower limit. I.e. any value below 1, as a invalid input data test case.

5. What is Integration testing ?

- System Integration Testing is testing between the 'System' and 'Acceptance' phases.
  - The System has already proven to be functionally correct, what remains to be tested is how the system reacts to other systems and/or organizations.
  - The objective of System Integration Testing is to provide confidence that the system or application is able to interoperate successfully with other specified software systems and does not have an adverse effect on other systems that may also be present in the live environment, or vice versa
  - It is possible that the testing tasks performed during System Integration Testing may be combined with System Testing, particularly if the system or application has little or no requirement to interoperate with other systems
  - In terms of the V Model, Systems Integration Testing corresponds to the Functional and Technical Specification phases of the software development lifecycle
  - Having completed Component integration testing and Systems testing, one must execute the plan for system-to-system integration
  - Infrastructure may need to be transformed in order to feed to an external system
  - Black Box testing techniques used.

6. what determines the level of risk ?

- A properly designed test that passes, reduces the overall level of Risk in a system
- • Risk – 'A factor that could result in future negative consequences; usually expressed as impact and likelihood'
- • When testing does find defects, the Quality of the software system increases when those defects are fixed
- • The Quality of systems can be improved through Lessons learned from previous projects
  • Analysis of root causes of defects found in other projects can lead to Process Improvement
  • Process Improvement can prevent those defects reoccurring
  • Which in turn, can improve the Quality of future systems
  • Testing should be integrated as one of the Quality assurance activities

7. What is alpha testing ?

- Type of testing a software product or system conducted at the developer's site. Usually it is performed by the end user.
  • It is always performed by the developers at the software development site.
  • Sometimes it is also performed by Independent Testing Team.
  • Alpha Testing is not open to the market and public
  • It is conducted for the software application and project.
  • It is always performed in Virtual Environment.
  • It is always performed within the organization.

- It is the form of Acceptance Testing
- Alpha Testing is definitely performed and carried out at the developing organizations location with the involvement of developers.
- It comes under the category of both White Box Testing and Black Box Testing.
- During this phase, the following will be tested in the application:
- Spelling Mistakes
- Broken Links
- Cloudy Directions
- Alpha Testing is always performed at the time of Acceptance Testing when developers test the product and project to check whether it meets the user requirements or not.
- It is always performed at the developer's premises in the absence of the users.
- It is considered as the User Acceptance Testing (UAT) which is done at developer's area.
- Unit testing, integration testing and system testing when combined are known as alpha testing.

8. What is beta testing ?

- It is always performed by the customers at their own site.
- It is not performed by Independent Testing Team.
- Beta Testing is always open to the market and public.
- It is usually conducted for software product.
- It is performed in Real Time Environment.
- It is always performed outside the organization.
- It is also the form of Acceptance Testing.
- Beta Testing (field testing) is performed and carried out by users or you can say people at their own

locations and site using customer data.

- It is only a kind of Black Box Testing.
- Beta Testing is always performed at the time when software product and project are marketed.
- It is always performed at the user's premises in the absence of the development team.
- It is also considered as the User Acceptance Testing (UAT) which is done at customers or users area.
- Beta testing can be considered "pre-release" testing.
- Pilot Testing is testing to product on real world as well as collect data on the use of product in the classroom.

9. What is component testing ?

- Component testing, also known as program or module testing, is done after unit testing.

  In this type of testing those test objects can be tested independently as a component without integrating with other components

  e.g. modules, classes, objects, and programs. This testing is done by the development team.

10. What is functional system testing?

- Functional Testing is a type of Software Testing in which the system is tested against the functional requirements and specifications.

  Functional testing ensures that the requirements or specifications are properly satisfied by the application.

11. What is Non-Functional Testing?

- Non-Functional Testing is defined as a type of Software testing to check non-functional aspects (performance, usability, reliability, etc) of a software application.

12. What is GUI Testing?

- GUI Testing is a software testing type that checks the Graphical User Interface of the Software.
- The purpose of Graphical User Interface (GUI) Testing is to ensure the functionalities of software application work as per specifications by checking screens and controls
- menus, buttons, icons, etc.

13. What is Adhoc testing?

- Ad hoc Testing is an informal or unstructured software testing type that aims to break the testing process in order to find possible defects or errors at an early possible stage.
- Ad hoc testing is done randomly and it is usually an unplanned activity which does not follow any documentation and test design techniques to create test cases.

14. what is load testing ?

- Testing technique that puts demand on a system or device and measures its response. It is usually conducted by the performance engineers.

15. What is stress testing ?

- Testing technique which evaluates a system or component at or beyond the limits of its specified

requirements. It is usually conducted by the performance engineer.

16. What is white box testing and list the types of white box testing?

- White box testing is software testing technique in which internal structure, design and coding of software are tested to verify flow of input – output and to improve
- Design, usability and security.
- in white box testing, code is visible to testers so it is aiso called clear box testing, open box testing, transparent box testing, Code-based testing and Glass box testing.
- Type :-

1. Branch Condition Testing
2. Branch Condition Combination Testing
3. Modified Condition Decision Testing
4. Dataflow Testing
5. Linear Code Sequence And Jump Testing

17. What is black box testing? What are the different black box testing techniques?

- a black box test, a tester will interact with the system's user interface by providing inputs and examining outputs without knowing how and where the inputs are worked upon.
- Testing techniques
  - Equivalence partitioning
  - Boundary value analysis
  - Decision tables
  - State transition testing
  - Use-case Testing
  - Other Black Box Testing

- Syntax or Pattern Testing

18. Mention what are the categories of defects?

- Common Types of Defects
  - Arithmetic Defects
  - Logical Defects.
  - Syntax Defects.
  - Multithreading Defects.
  - Interface Defects.
  - Performance Defects.

19. Mention what bigbang testing is?

- Testing technique which integrates individual program modules only when everything is ready. It is performed by the testing teams.

20. What is the purpose of exit criteria?

- The purpose of exit criteria to prevent a task from being considered completed when there are still outstanding parts of the task which have not been finished . Exit criteria are used to report against and to plan when to stop testing.

21. When should "Regression Testing" be performed?

- Type of software testing that seeks to uncover software errors after changes to the program (e.g. bug fixes or new functionality) have been made, by retesting the program. It is performed by the testing teams.

22. What is 7 key principles? Explain in detail?

1Testing shows presence of Defects

2 Exhaustive Testing is Impossible!

3 Early Testing.

4 Defect Clustering

5 The Pesticide Paradox

6 Testing is Context Dependent

7 Absence of Errors Fallacy

1. **TESTING SHOWS PRESENCE OF DEFECTS**

- Testing can show that defects are present, but cannot prove that there are no defects.
- Testing reduces the probability of undiscovered defects remaining in the software but, even if no defects are found, it is not a proof of correctness.
- We test to find Faults
- As we find more defects, the probability of undiscovered defects remaining in a system reduces.

2. **EXHAUSTIVE TESTING IS IMPOSSIBLE!**

- Testing everything including all combinations of inputs and preconditions is not possible.
- So, instead of doing the exhaustive testing we can use risks and priorities to focus testing efforts.
- For example: In an application in one screen there are 15 input fields, each having 5 possible values, then to test all the valid combinations you would need 30 517 578 125 (515) tests.
- That is we must Prioritise our testing effort using a Risk Based Approach.

3. **EARLY TESTING**

- Testing activities should start as early as possible in the software or system development life cycle, and should be focused on defined objectives.
- Testing activities should start as early as possible in the development life cycle

4. **DEFECT CLUSTERING**

• A small number of modules contain most of the defects discovered during pre-release testing, or are responsible for the most operational failures.

• Defects are not evenly spread in a system They are 'clustered'

5. **PESTICIDE PARADOX**

If the same tests are repeated over and over again, eventually the same set of test cases will no longer find any new defects.

To overcome this "pesticide paradox", the test cases need to be regularly reviewed and revised, and new and different tests need to be written to exercise.

different parts of the software or system to potentially find more defects.

6. **TESTING IS CONTEXT DEPENDENT**

. Testing is basically context dependent, Testing is done differently in different contexts

. Diffrenent kinds of sites are tested differently.

7. **ABSENCE OF ERRORS FALLACY**

If the system built is unusable and does not fulfill the user's needs and expectations then finding and fixing defects does not help.

If we build a system and, in doing so, find and fix defects – It doesn't make it a good system.

Even after defects have been resolved it may still be unusable and/or does not fulfil the users.

23. Difference between QA v/s QC v/s Tester

- While QA testing focuses on providing assurance that quality requested will be achieved, QC testing focuses on fulfilling the quality requested. QA focuses on
- preventing defect while QC focuses on identifying the defect.

24. Difference between Smoke and Sanity?

- Smoke testing is a subset of acceptance testing whereas Sanity testing is a subset of Regression Testing. Smoke testing is documented or scripted whereas

- Sanity testing isn't. Smoke testing verifies the entire system from end to end whereas Sanity Testing verifies only a particular component.

25. Difference between verification and Validation

- **Verification :**

  It includes checking documents, design, codes and programs.
  Verification is the static testing.
  It does not include the execution of the code.
  Quality assurance team does verification.
  It comes before validation.
  It consists of checking of documents/files and is performed by human.

- **Validation :**

  It includes testing and validating the actual product.
  Validation is the dynamic testing.
  It includes the execution of the code.
  Validation is executed on software code with the help of testing team.
  It comes after verification.
  It consists of execution of program and is performed by computer.

26. Explain types of Performance testing.

- Load Testing :

27. What is Error, Defect, Bug and failure?

- A mistake in coding is called Error, error found by tester is called Defect, defect accepted by development team

then it is called Bug, build does not meet the requirements then it Is Failure.

28. Difference between Priority and Severity

- Severity is a parameter to denote the impact of a particular defect on the software. Priority is a parameter to decide the order in which defects should be fixed.

  Severity means how severe defect is affecting the functionality. Priority means how fast defect has to be fixed.

29. What is Bug Life Cycle?

- Bug life cycle also known as defect life cycle is a process in which defect goes through different stages in its entire life. This lifecycle starts as soon as a bug is reported by the tester and ends when a tester ensures that the issue is fixed and won't occur again.

30. Explain the difference between Functional testing and NonFunctional testing

- **Functional Testing**
- Functional testing is performed using the functional specification provided by the client and verifies the system against the functional requirements.
  Functional testing is executed first
  Manual testing or automation tools can be used for functional testing
  Business requirements are the inputs to functional testing
  Functional testing describes what the product does
  Easy to do manual testing
  Types of Functional testing are
  - Unit Testing
  - Smoke Testing

- Sanity Testing
- Integration Testing
- White box testing
- Black Box testing
- User Acceptance testing
- Regression Testing

➢ **Non-Functional Testing**

➢ Non-Functional testing checks the Performance, reliability, scalability and other non-functional aspects of the software system.
Non functional testing should be performed after functional testing
Using tools will be effective for this Testin
Performance parameters like speed , scalability are inputs to non-functional testing.
Nonfunctional testing describes how good the product works
Tough to do manual testing
Types of Nonfunctional testing are

- Performance Testing
- Load Testing
- Volume Testing
- Stress Testing
- Security Testing
- Installation Testing
- Penetration Testing
- Compatibility Testing
- Migration Testing

31. What is the difference between the STLC (Software Testing Life Cycle) and SDLC (Software Development Life Cycle)?

- **SDLC :**
- SDLC is mainly related to software development.
  Besides development other phases like testing is also included.
  SDLC involves total six phases or steps.
  In SDLC, more number of members (developers) are required for the whole process.
  In SDLC, development team makes the plans and designs.
  Goal of SDLC is to complete successful development of software.
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  It helps in developing good quality software.
  SDLC phases are completed before the STLC phases.
- **STLC** :
- STLC is mainly related to software testing.
  It focuses only on testing the software.
  STLC involves only five phases or steps.
  In STLC, less number of members (testers) are needed.
  In STLC, testing team makes the plans and designs.
  Goal of STLC is to complete successful testing of software.
  It helps in making the software defects free.
  STLC phases are performed after SDLC phases.

32. What is the difference between test scenarios, test cases, and test script?

- A test scenario is a high-level description of what to test.
  Scenario: What to test
- A test case provides detailed steps and expected results.

Test Case: How to test

- A test script is an automated code used to execute test cases. Test Script: Code to test

33. Explain what Test Plan is? What is the information that should be covered ?

- A Test Plan is a document that outlines the strategy, scope, and schedule of testing activities. It includes details like objectives, scope, test environment, schedule, roles, risks, and entry/exit criteria.

34. What is priority?

- Priority indicates how quickly a defect should be fixed based on its importance to the business. It helps determine the order in which bugs are resolved.

35. What is severity?

- Severity is the degree of impact a defect has on the system's functionality. It helps determine how serious the bug is from a technical perspective.

36. Bug categories are…

- **Bug categories** are different types of errors found in software or applications.
- **Main Bug Categories**:
- **Functional Bug**
  When a feature does not work as expected.
  Example: Login button not working.
- **Performance Bug**
  When the system is slow or crashes.
  Example: Website takes too long to load.

- **UI (User Interface) Bug**
  Issues related to design or layout.
  Example: Text is cut off or button is misplaced.
- **Security Bug**
  When there is a risk to data safety.
  Example: Passwords are not stored securely.
- **Compatibility Bug**
  When software does not work properly on different devices or browsers.
  Example: Works on mobile but not on desktop.
- **Logical Bug**
  When the program logic is incorrect.
  Example: Wrong calculation result.

37. Advantage of Bugzila .

1. Open Source and Free
Bugzilla is free to use, which reduces project cost.
2. Easy Bug Tracking
Helps in tracking, managing, and updating bugs efficiently throughout their life cycle.
3. Customizable
Can be customized according to project needs (fields, workflows, reports).
4. Email Notifications
Automatically sends email updates for bug status changes.
5. Strong Search and Reporting
Provides powerful search options and detailed reports for defects.
6. User Access Control
Allows role-based access to maintain security and control.
7. Integration Support
Can be integrated with other tools for better workflow.

38. Difference between priority and severity ?

- Severity is a parameter to denote the impact of a particular defect on the software. Priority is a parameter to decide the order in which defects should be fixed.

Severity means how severe defect is affecting the functionality. Priority means how fast defect has to be fixed.

39. What are the different Methodologies in Agile Development Model?

- Agile development includes methodologies like Scrum, Kanban, Extreme Programming (XP), Lean, Crystal, and Feature-Driven Development (FDD). These methods focus on iterative development, collaboration, and flexibility.

40. Explain the difference between Authorization and Authentication in Web testing.What are the common problems faced in Web testing?

- Authentication verifies user identity, while authorization controls access to resources.
- Common web testing issues include browser compatibility, performance, security, UI/UX problems, session management, and broken links.

41. When to used Usablity Testing?

- Usability testing is used to evaluate how easy and user-friendly a system is for users. It is performed during design, after development, and before release to improve user experience.

42. What is the procedure for GUI Testing?

- GUI testing involves verifying the user interface for layout, design, and functionality. The procedure

includes understanding requirements, creating test cases, validating UI elements, checking usability, and reporting defects.